

Rough edges in embedded Linux

When Linux is ported to an embedded or handheld device, it has to be modified to suit the low memory, low horsepower and low resource environments of the embedded world. The task of removing unwanted components, configuring the kernel to use minimal resources and reducing the size is a complex one.

There is also a legacy issue in Linux that impacts user experience: The boot and shutdown operations are carryovers of the original System V UNIX system and they were not written with short boot time or a smaller memory in mind. To get Linux adopted in the embedded world, the sources need to be modified to reduce the boot time. The user experience will be affected if devices like mobile phones take a long time to boot.

There is a dire need to change this structure to make Linux boot and shutdown faster. We will look at different approaches to this problem and the work being done in this area.

The virtualisation approach

Since 2006, the x86 (as well as x86-64) breed of processors also started featuring hardware-assisted virtualisation and, over the years, have become quite popular on laptops and netbooks. In this approach, a hardware feature of x86 architecture (dubbed Intel VT or AMD-V) is used.

A thin hypervisor is first started. This is typically based on Linux, and it boots in 5 to 6 seconds flat. It is a bare-metal virtualiser that brings up four or five popular applications like a browser, IM client, e-mail client, Skype and so on. Meanwhile, the boot loader of Windows (or any other full-fledged OS) is started in the background. If users want to go to Windows mode, they can press a key and switch to Windows and a fast-boot environment.

The popular products in the category include Splashtop from DeviceVM and Hyperspace from Phoenix technologies. Other products that use these technologies include HP's QuickWeb series of laptops, Asus Expressgate, Acer and so on.

The optimisation approach

When the virtualisation-based solution is not possible, the only other option left is to optimise the Linux boot

and shutdown code to reduce the time.

In the original System V UNIX system, the boot and shutdown are handled by user-level scripts located in the `/etc/rc/` directories. These scripts poll for devices, create nodes for the devices, start services, and the code is written in shell scripts. They spawn many processes and take a big toll on the memory and performance. One method is to start many of the unconnected services in parallel. The other is to remove the scripts and rewrite them as a single application.

Moving from glibc to ulibc also helps, as it is a leaner library. Many unwanted modules and BUS snoops can be cut down to reduce time. For example, if the embedded system does not have any IDE, the IDE probe code can be removed completely. Writing on the console is avoided to save time. All these changes involve a study of the system, careful tuning of the parameters and kernel code, and observing the results. Many tricks and techniques for optimisation are documented in http://elinux.org/Boot_Time

Besides the above-mentioned techniques, some companies are coming up with optimised distributions to cut the booting time.

- Montavista has announced an optimised version of embedded Linux on the Freescale platform that boots in one second.
- Moblin, promoted by Intel for the MID/netbook market and which is based on ATOM processors, boots in five seconds for embedded Linux.

As Linux is getting into mainstream consumer devices like netbooks, mobile phones, set-top boxes, etc, there is a need to drastically speed up the boot time. And, at the end of the day, efforts that address this factor will help in improving the overall customer experience by cutting down the long waits in boot and shutdown operations. 

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